

Introduction to Software Engineering

COSC 336

ReSource (Phase II)

Feasibility Study

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle
Management System

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Introduction

Concept

ReSource is an intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System designed to support University's asset management department in identifying, sharing, and reuse extra/underutilized assets such as furniture, equipment, vehicles, and books. By doing so a department that requests an asset could use another department's surplus asset and avoid wasteful spending and storage. Furthermore, the system is designed to track each asset's life-cycle from its registration, maintenance, transfer, and finally when it's donated/recycled or disposed of. The system combines traditional asset management with AI features such as semantic matching, automated classification, and sustainability driven recommendations.

Document Purpose

This document's purpose is to serve as the phase 1 deliverable which highlights the initial project plan and requirement gathering for ReSource. This document will be the foundation of our project that we will turn to as a reference point to verify that we meet all the requirements stated.

Project Overview

High-Level System Description

ReSource is foundationally built on two layers, a traditional asset management layer and an AI-Powered layer.

Layer 1 Classic Asset Management

Module	Function
Asset Registration	Captures structured asset data such as name, category, condition, quantity, location, estimated value, and images
Resource	Publishes registered assets as searchable/filterable listings visible to other

Marketplace	departments
Request & Reservation	Lets a department submit a request for an asset specifying purpose, quantity, and urgency
Approval & Transfer	Routes accepted requests to an administrator on approval, updates ownership, custody, and location
Life-Cycle Tracking	Logs maintenance, repair, donation, recycling, and disposal events in the asset's history
Access Management	Assigns role-based permissions, governing which actions each user can perform

Layer 2 AI-Powered Features

Module	Function
AI Matching Engine	Ranks open requests against available assets using semantic similarity plus condition, location, and urgency as inputs
AI Classification	Suggests a category and tags at the point of registration, based on the asset's description
Sustainability Optimizer	Outputs a recommended end-of-life action for the asset whether its reuse, repair, donate, recycle, or dispose
LLM Assistant	Accepts natural language questions and maps them to a search/request action
Generative Reporting	Produces a text summary of a selected time period's activity like transfers, savings, and waste diversions from the dataset

Background and Context

Universities routinely purchase new furniture, computers, laboratory equipment, and office materials, while similar usable items remain unused, stored, or thrown out by other departments. This is often not caused by a shortage of resources, but by the absence of an integrated system to identify, match, and track them.

Nowadays, when a department has surplus assets or needs resources, it's typically handled informally and without a shared system. A department that no longer needs an asset stores it in a stockroom, reports it to facilities for disposal, or occasionally informs other departments by email or word of mouth, without any central listing of what is available. When a department needs a certain asset, it will make a request without checking if another department already has something suitable in surplus. Approvals for transferring assets,

when they happen at all, are done through emails or paper forms, with no standardized workflow. Maintenance and repair records, when kept, are usually stored in each department's own spreadsheets or paper logs, making it difficult to see an asset's full history. Decisions about whether to donate, recycle, or dispose of an asset are made case by case, with no consistent criteria and no tracking of the financial and environmental impact.

This manual process of managing assets is unreliable and wastes a significant amount of time and money that could be saved through better coordination. ReSource is designed to replace this outdated, wasteful process with a single, structured platform.

Business Goals and Objectives

- Reduce unnecessary purchases by connecting departments with existing surplus assets.
- Extend the life of campus assets through better tracking and maintenance.
- Increase reuse, donation, and recycling rates across the university.
- Provide insights into the environmental and financial impact of the circular resource management.
- Demonstrate how classic and AI-enhanced features can be combined in a single system.

Expected Benefits

- Financial: Reduced spending by reusing surplus assets and lowering storage costs.
- Operational: clearer accountability through approval workflows and easy to review existing assets.
- Environmental: Reduced waste through donations, recycling, and reusing of assets.
- Organizational: A single source for asset location, condition, and history replacing scattered and informal communication.

Assumptions

- The university would be willing to adopt a centralized platform for internal asset exchange if one were available.
- Departments are willing to accurately register and update the status of their assets.
- Users will have basic computer literacy sufficient to use a web-based system without extensive training.

Project Constraints

- Timeline: The project must be completed within the Fall 2026 semester, following the official Phase 1-8 schedule (September to December 2026).
- Team size: The project is developed by a team of 4 students, with work distributed across planning, requirements, design, and development tasks.
- Scope: As an academic prototype, the system will demonstrate core functionality rather than a production-ready, university-wide deployment.

- Technical resources: AI features (matching, classification) will rely on existing libraries, pretrained models, or APIs appropriate for a semester-length project, rather than custom trained machine learning models.
- Data availability: Real University asset data is not accessible for this course project, so sample or simulated data will be used for development and testing.

Project Scope

In scope	Out of scope
Core classic modules: asset registration, resource marketplace, request/reservation, approval and transfer workflows, life-cycle history, maintenance/repair logging, and role-based user management	Integration with real university procurement or financial systems
Core AI-enhanced modules: AI-assisted classification, AI-based semantic matching with a basic explanation of recommendations, and a sustainability recommendation feature	Full production-level security certification or compliance auditing
Basic operational and AI-generated reporting	Large-scale, continuously trained AI models; AI components will use existing tools/APIs rather than custom model training

note: a basic natural-language query feature may be included as time allows, but is not a core deliverable

Stakeholders

As stated above, the system will support campus resource reuse, asset tracking, and sustainable decisions. This project involves many users with different authorizations and access limits. Therefore understanding their responsibilities and expectations help in defining suitable features, information access, and authorization control.

The table below identifies all the possible stakeholders, starting from the system developing team to all the different end-users. The table clearly states their roles/responsibilities and their needs/expectations to help build up the system.

Stakeholder (Category)	Role/Responsibility	Needs/Expectations & Data Needs	Feature Requirement Mapping	Permissions & Approval Decisions
Administrators (User)	Manage operational workflows, resolve allocation conflicts, approve transfers or end-of-life actions	Approval queues, operational reports, records of rules and changes; needs visibility into all pending requests and transfer records	Final Approval and Transfer Management with User and Access Management	Approve/reject asset transfers, end-of-life actions, and manage user roles
Procurement Team (User)	Check stock before purchasing and accept or reject requests	Inventory and demand visibility, compatibility information, purchase records; needs inventory levels and purchase-request history	AI Matching Engine giving AI-Generated Reports	View inventory and requests and approve whether a new purchase is justified or if we can reuse surplus assets
Department Representative (User)	Identify department needs and surplus assets with coordinated requests and transfers	Asset inventory and request status since they need visibility into their own department's inventory and requests	Request & Reserve New Asset	Create/edit own department's listings and submit requests while approval sits with Administrators
Asset Coordinators (User)	Register assets and maintain condition, ownership, and location records plus verify availability and review suggested matches	Accurate asset histories, current availability, clear specifications, relevant matches, status updates	Asset Registration & Publication, AI Matching, Sustainability Optimization Resource Marketplace, and Life-Cycle Tracking	Register/update assets and confirm receipt/condition on completed transfers
University Management (Customer)	Represent the university, confirm project priorities and budget authority, sign off on major decisions	Accurate asset records, reduced spending, clear high-level reporting with needs summary-level cost and impact data, not operational detail	Generative AI Reporting (financial savings, waste diversion, avoided purchases)	Final authority on budget and adoption without day-to-day operational permissions
Developers (Software Developer)	Gather/understand requirements, design, build, and test the system	Clearly stated requirements and scope, defined interfaces, access to development tools/environment	Not feature-specific because they are responsible for implementing all classic and AI modules	Full read/write access to codebase and dev environment without business-decision approval authority

Lab Instructors (Development Manager)	Provide academic guidance, review phase deliverables, assess progress against course requirements	Regular, clear deliverables and documentation visibility into evaluation criteria and rubric alignment across phases	Not feature-specific since they governs project process and academic evaluation, not system functionality	Approve/grade phase deliverables without permissions within the deployed system itself
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The Owner of the system is the university head of facilities management
And the Technical owner is the IT department

These stakeholder needs will guide the system’s development and evaluation.

Market Analysis

Technical Analysis

1) Hardware and Software Requirements

ReSource is proposed as a web-based system that combines campus asset management with AI-assisted resource matching, classification, and sustainability recommendations. Development will use free and open-source tools, pretrained models, or APIs, consistent with the scope defined in Phase 1.

- **Backend (Python/Django or Java):** The backend would process asset registrations, requests, reservations, approvals, transfers, and maintenance records. Django provides user authentication of accounts, groups, permissions, and password hashing [1].
- **Frontend (HTML, CSS, and JavaScript):** The interface would provide asset listings, search filters, registration forms, request tracking, and dashboards appropriate to each user's role.
- **Database (MySQL) :** Database would store users, departments, assets, requests, transfers, and life-cycle records. Linking these records would support ownership tracking and reporting. MySQL is supported by Django as well.
- **AI Libraries:** Use pre-defined AI libraries such as (Sentence Transformers) for semantic search, comparing asset and description. LLM could also be used to support generated reports and natural-language interaction
- **Hardware:** Existing teams' and labs computers would be used for development and testing.

For example, when a department needs equipment, a user would enter the request through the frontend (HTML, CSS, and JavaScript). The backend (Python/Django or Java) would process the request and retrieve available asset records from the MySQL database. Sentence Transformers would compare the request's meaning with asset descriptions to help identify suitable matches. The backend would apply checks for availability and required specifications, and the frontend would display the results.

When the user selects an asset, the backend would verify permissions and quantities, record the reservation in MySQL, and route the request for approval. After the transfer is confirmed, it would update the asset's custody, location, and life-cycle history in the database. An LLM could subsequently generate a summary using the recorded transfer information.

2) Skills Required

The project requires the following steps:

- **Software Engineering skills:** Designing/Planning, Implementing, and testing skills
- **Web Development:** UI, and backend workflows. Making the system easy to use.
- **Database Design:** Connecting assets, user, request, and maintenance records while maintaining accurate quantities and ownership.
- **AI integration** (layer 2 of the system): Using pretrained models or APIs, preparing input data, evaluating recommendations, and handling unsuccessful responses.
- **Collaboration and deployment:** Managing shared code through GitHub, integrating modules, and configuring the demonstration environment.

For example: One team member develops the interface, another builds the backend and approval workflows, a third designs the database for asset and transfer records, and the fourth integrates AI matching and reporting. All four members contribute to testing and combining the modules.

3) Feasibility

- With these tools and skills, the project is possible within the proposed timeline
- Open-source tools can reduce licensing costs
- Team members already have the basic skills needed, and extra parts (like AI) can be learned during the project.

As real asset data is unavailable, development and evaluation will use sample or simulated records. Testing will follow the Phase 1 success criteria, including the planned workload of 100 asset records and five simultaneous users. AI accuracy and response-time targets remain to be demonstrated.

Overall, ReSource appears technically feasible as a prototype, provided the team confirms the required skills, validates the selected AI tools, and maintains the agreed scope.

Technical Ownership

The IT Department would be the technical owner of ReSource, responsible for hosting, technical maintenance, security updates, backups, and support in a future university deployment. During the academic prototype, the student team would manage the development and demonstration environment.

Data & AI

Data Preparation and Quality

The sample dataset described above will include asset descriptions, categories, specifications, quantities, condition, location, and availability, together with related request and maintenance records. The team will check missing fields, duplicate identifiers, inconsistent categories, and invalid quantities before using these records for matching or reporting. Missing information will be flagged rather than treated as verified data.

The prototype will use synthetic campus asset records generated with AI assistance and based on clearly stated assumptions. These records will represent assets such as furniture, computers, and laboratory equipment, with realistic descriptions, quantities, conditions, and locations. The team will review the records for completeness and consistency and verify expected categories and suitable matches before evaluation. The dataset will support development and testing and will not represent actual university inventory.

AI Contribution and Team Contribution

Pretrained AI components would provide semantic comparison, category suggestions, and generated summaries. The team's contribution will be to prepare the data, integrate these components into ReSource, define matching and approval rules, and evaluate the results. Classification suggestions will require user review, while sustainability recommendations will depend on recorded condition, maintenance information, available costs, and approved decision criteria.

The backend will enforce permissions, availability, quantities, and approvals. It will also calculate report totals, which an LLM may summarize. This separation will allow AI to support decisions while keeping operational controls and numerical calculations within the application.

The main technical and AI risks are:

- Incomplete data: Any record missing a required matching field will be flagged for correction before matching.
- Inaccurate recommendations: Results below the matching or classification targets will trigger review of the data and selected model. Manual category selection and structured search will remain available.
- Slow or unavailable AI: At least 90% of at least 20 AI interactions should return usable results within 15 seconds. Timeouts or service failures will provide a clear fallback to ordinary forms, structured search, or conventional reports.

Financial Analysis

Legal and regulatory analysis

Risk assessment

Methodological basis. The risk assessment was structured using established risk-management guidance. NIST SP 800-30 describes risk assessment as a process that includes preparing for the assessment, conducting the assessment, and maintaining the assessment over time [2]. ISO 31000 provides a broader organizational framework built around identifying, analyzing, evaluating, treating, monitoring, and communicating risk [3]. These sources were used to shape the process, while the numerical ratings below are project-specific estimates made for this feasibility study rather than values supplied by NIST or ISO.

Project scoring method. For each identified risk, the team assigns a Likelihood score from 1 to 5 and an Impact score from 1 to 5. The overall score is calculated as Risk Score = Likelihood x Impact. This 1-5 scoring model and the thresholds below are an adaptation for this project so that risks can be compared consistently.

Score	Rating	Interpretation
1	Very Low	Unlikely / minimal effect
2	Low	Possible but limited effect
3	Medium	Reasonably possible / noticeable effect
4	High	Likely / serious effect
5	Very High	Very likely / severe effect

Risk Score	Priority
1-5	Low
6-10	Moderate

11-15	High
16-25	Critical

Source-to-risk mapping. AI-related risks were informed by the NIST AI Risk Management Framework, which emphasizes trustworthiness characteristics such as validity and reliability, security and resilience, accountability and transparency, explainability and interpretability, privacy enhancement, and fairness with harmful bias managed [4]. Generative-AI and LLM risks were further informed by the NIST Generative AI Profile [5] and the OWASP Top 10 for LLM Applications 2025 [6]. General web-application security risks were informed by the OWASP Top 10:2025 [7], while recommended security and privacy controls were aligned at a high level with NIST SP 800-53 Rev. 5 [8].

2. Risk Register

The following register applies the methodology above to the proposed semester prototype. Mitigation describes actions intended to reduce likelihood or impact before a risk occurs. Contingency describes the fallback action if the risk materializes.

ID	Risk	L	I	Score	Mitigation	Contingency	Basis
R1	Insufficient or poor-quality asset/request data for AI functions	4	5	20 - Critical	Define required fields early; validate completeness and consistency; clean records; use representative test data.	If suitable real data is unavailable, use a controlled synthetic/demo dataset and clearly state the limitation.	[1], [3], [4]
R2	AI classification or semantic matching produces inaccurate recommendations	4	4	16 - Critical	Test against predefined examples; track confidence; compare AI output with deterministic rules; require human review.	Use conventional category, keyword, filter, or rule-based matching when AI confidence is low or the service fails.	[3], [4]
R3	AI/LLM service or API integration fails or becomes unavailable	3	4	12 - High	Separate AI features behind clear service interfaces; test integrations early; define error handling and timeouts.	Use local/simple algorithms or mocked responses so the core prototype remains demonstrable.	[4], [5]
R4	Project scope is too large for the semester timeline	4	5	20 - Critical	Prioritize essential workflows; use incremental development; assign clear owners; review progress weekly.	Defer lower-priority enhancements while preserving a complete baseline workflow and core AI demonstration.	[1], [2]
R5	Integration problems between modules developed by different team members	3	4	12 - High	Define interfaces and shared data contracts early; integrate frequently; use code review and version control.	Temporarily freeze unstable features and integrate a simpler working implementation.	[1], [2]
R6	Unauthorized access or exposure of user, asset, financial, or departmental information	3	5	15 - High	Use role-based access control, secure authentication, least privilege, data minimization, protected storage, and audit logging.	Disable affected functions, rotate credentials, restore safe configurations, and use non-sensitive demo data until corrected.	[6], [7]
R7	Bias or unfair prioritization in AI-supported matching or recommendations	3	4	12 - High	Use transparent factors; test for systematic differences; document criteria; retain authorized human review.	Allow authorized overrides and require manual approval for consequential allocation decisions.	[3]

R8	AI recommendations cannot be adequately explained to users	3	4	12 - High	Display the main factors behind each recommendation, relevant confidence information, and decision rationale.	Route low-explainability or disputed recommendations to manual review instead of automatic action.	[3], [4]
R9	Web or application security vulnerabilities affect the prototype	3	5	15 - High	Apply secure authentication, access control, input validation, safe configuration, dependency management, and logging.	Disable compromised components, patch the issue, rotate secrets, and operate only unaffected modules until recovery.	[6], [7]
R10	Prompt injection or sensitive-information disclosure through the LLM assistant	3	5	15 - High	Limit accessible data/tools; validate inputs and outputs; separate trusted instructions from user content; avoid exposing secrets.	Disable the LLM feature or restrict it to non-sensitive demo data until controls are corrected.	[5], [4]
R11	Team member availability or uneven workload causes delays	3	4	12 - High	Assign primary and backup owners; maintain shared documentation; review workload and blockers regularly.	Redistribute critical tasks and reduce nonessential scope if resource availability declines.	[1], [2]
R12	Unexpected infrastructure, hosting, or AI-service cost	2	3	6 - Moderate	Use free/student tiers where appropriate; cap API usage; prefer low-cost or local tools during prototyping.	Switch to local/open-source or lower-cost alternatives and reduce AI calls to demonstration scale.	[1], [2]
R13	Sustainability or carbon-saving estimates are misleading because assumptions/data are weak	3	3	9 - Moderate	Document assumptions, formulas, data sources, and uncertainty; validate calculations with known examples.	Present results as estimates/ranges and suspend unsupported indicators until better data is available.	[3], [4]
R14	Users find proposed workflows difficult to adopt	2	4	8 - Moderate	Keep workflows simple; test mockups with representative users; align permissions and steps with real roles.	Simplify approval/request flows and provide guided instructions or a reduced workflow for the prototype.	[1], [2]

Interpretation of the highest scores. The two highest project-specific risks are data availability/quality (R1) and semester scope/time (R4), both scored 20. AI recommendation accuracy (R2) is scored 16. These values reflect the team's feasibility judgement after applying the stated method; they are not published ratings from the cited standards.

3. Project Schedule Feasibility

Schedule feasibility was evaluated by comparing the amount of work required with the semester milestones, team availability, integration dependencies, and the need to validate AI features before they become dependencies for the rest of the system. Risk-management guidance supports reviewing risks as conditions change rather than treating assessment as a one-time activity [2], [3].

Stage	Timing	Internal Target
Phase 2 - Feasibility	Current phase	Finalize feasibility evidence, quantified risk register, schedule assessment, and recommendations.
Phase 3 - Requirements	Next milestone	Finalize functional/non-functional requirements, actors, permissions, AI requirements, data requirements, and acceptance criteria.

Phases 4-5 - Architecture & Design	Design milestone	Define application/database/AI/security architecture, interfaces, UML/ER models, UI mockups, and fallback behavior.
Phase 6 - Development	Implementation milestone	Build baseline modules first, then integrate AI classification/matching, recommendation explanations, sustainability indicators, and reports.
Phase 7 - Testing	QA milestone	Run unit, integration, workflow, security, AI-quality, and user-acceptance tests; fix high-priority defects.
Phase 8 - Final Submission	Final milestone	Complete documentation, demonstration preparation, final integration, and presentation.

4. Schedule Risk Controls

Prioritize the baseline workflow first. Authentication, assets, requests, search, approval, transfer, and maintenance should work before advanced AI enhancements are expanded.

Prototype AI early. Run a small classification/matching proof of concept before the architecture depends on an external AI service.

Integrate continuously. Merge and test modules regularly so incompatibilities are discovered while they are still inexpensive to fix.

Track risks weekly. Update likelihood, impact, mitigation status, owners, and residual risk as evidence changes [2], [3].

Keep fallback paths. The prototype should retain deterministic search/filtering and manual review so AI failure does not block the entire system [4], [5].

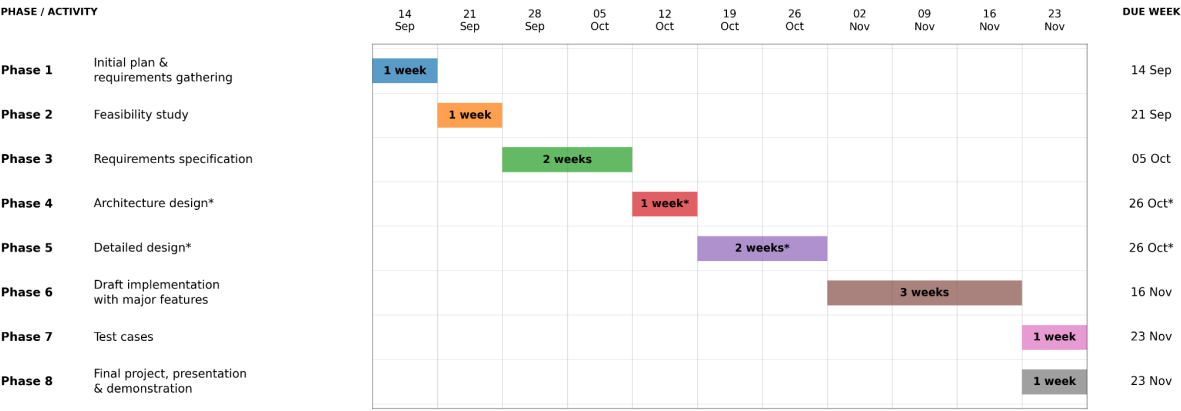
Project schedule

Phase	Main Activities	Milestone	Dependencies
Phase 1 Planning and Requirements Gathering	Define the scope, stakeholders, initial requirements, constraints, and preliminary project plan.	Initial plan and requirement gathering document completed. <i>Due: week of 14 Sep 2026</i>	None. Project initiation.
Phase 2 Feasibility Study	Assess technical, economic, operational, and schedule feasibility, including AI tools and data availability.	Feasibility document completed. <i>Due: week of 21 Sep 2026</i>	Phase 1: Scope and initial requirements.
Phase 3 Requirements Specification	Finalize functional, non-functional, AI, and sustainability requirements, business rules, and acceptance criteria.	Requirements document completed. <i>Due: week of 5 Oct 2026</i>	Phases 1 and 2: Stakeholder needs and feasibility findings.

Phase 4 Architecture Design	Define system layers, modules, interfaces, data flow, and the AI integration approach.	Architecture section completed for the combined design document. <i>Due: week of 26 Oct 2026</i>	Phase 3: Completed requirements specification.
Phase 5 Detailed Design	Specify database schemas, workflows, UML models, user interfaces, and component-level logic.	Combined architecture and detailed design document completed. <i>Due: week of 26 Oct 2026</i>	Phases 3 and 4: Requirements and system architecture.
Phase 6 Implementation	Develop core asset-management functions and planned AI features; integrate major system modules.	Draft implementation with major features completed. <i>Due: week of 16 Nov 2026</i>	Phases 4 and 5: Completed architecture and detailed design.
Phase 7 Testing and Test Cases	Prepare and execute functional, integration, usability, security, and AI test cases; document results and corrections.	Test-case details completed. <i>Due: week of 23 Nov 2026</i>	Phases 3 and 5: Test basis. Phase 6: Prototype for test execution.
Phase 8 Final Documentation and Delivery	Finalize documentation and corrections; prepare the project presentation and system demonstration.	Final project submitted; presentation and demonstration delivered. <i>Due: week of 23 Nov 2026</i>	Phases 6 and 7: Working prototype, test results, and necessary corrections.

ReSource | Project Gantt Chart

Waterfall development • Eight phases • Fall 2026



* Phases 4-5: proposed 1 + 2 week split. The combined design document is due in the week of 26 October.
Phases 7-8 share the final one-week window and are both due in the week of 23 November.
Source: supplied course schedule. Bars are proposed weekly allocations; exact start dates and lab-day deadlines are not specified.

Recommendations

Constraints

Constraint	Why it matters	Simple response
Legal	The system must follow applicable laws when handling university data and AI services.	Use only approved data and services; check legal requirements before deployment.
Ethical	AI recommendations should not unfairly favor departments or make important decisions by itself.	Keep humans responsible for approvals and allow users to review or override AI suggestions.
Privacy	Asset records may include user, department, location, financial, or operational information.	Collect only needed data and limit who can view sensitive information.
Security	Unauthorized access or changes could affect asset ownership, transfers, and records.	Use authentication, role-based access, secure storage, and audit logs.
Procurement	External AI tools, cloud services, or software may require university approval or payment.	Prefer approved tools and confirm cost, licensing, and vendor requirements early.
Ownership	The system tracks who owns or controls each asset and records transfers between departments.	Require authorized approval before changing ownership, custody, or location.
Safety	Some assets may be damaged, unsafe, or unsuitable for reuse or transfer.	Require inspection or maintenance checks before risky assets are reused or transferred.
University Policy	The project must follow university rules for IT, data, assets, procurement, and disposal.	Align workflows and permissions with university procedures and responsible staff.

AI Contributions

Technical analysis was prepared by proposing the hardware and software tools, and explaining how the frontend, backend, database, and AI components would work together. The required skills were identified as well, request and transfer workflow, and outlined the prototype's constraints and planning testing workload was completed by the team members.

AI supported this work by improving wording and organization and helping draft additions on data quality, privacy, technical risks, and evaluation. The team selected relevant suggestions for inclusion and remains responsible for checking the content, confirming assumptions, and making the final decisions. It was also used as a reference to compare the available approaches to use in building the interface and the backend.

References

- [1] "User authentication in Django | Django documentation," *Django Project*.
<https://docs.djangoproject.com/en/6.1/topics/auth/>
- [2] R. Ross, "Guide for Conducting Risk Assessments," NIST Special Publication 800-30 Rev. 1, National Institute of Standards and Technology, Gaithersburg, MD, USA, Sep. 2012, doi: 10.6028/NIST.SP.800-30r1.
- [3] International Organization for Standardization, ISO 31000:2018, Risk Management - Guidelines, 2nd ed. Geneva, Switzerland: ISO, 2018.
- [4] E. Tabassi, Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1, National Institute of Standards and Technology, Gaithersburg, MD, USA, Jan. 2023, doi: 10.6028/NIST.AI.100-1.
- [5] C. Autio, R. Schwartz, J. Dunietz, S. Jain, M. Stanley, E. Tabassi, P. Hall, and K. Roberts, Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile, NIST AI 600-1, National Institute of Standards and Technology, Gaithersburg, MD, USA, Jul. 2024, doi: 10.6028/NIST.AI.600-1.
- [6] OWASP Foundation, "OWASP Top 10 for LLM Applications 2025," Nov. 17, 2024. [Online]. Available: <https://genai.owasp.org/resource/owasp-top-10-for-llm-applications-2025/>. [Accessed: Sep. 29, 2026].
- [7] OWASP Foundation, "OWASP Top 10:2025," 2025. [Online]. Available: <https://top10.owasp.org/2025/>. [Accessed: Sep. 29, 2026].
- [8] R. Ross and V. Pillitteri, "Security and Privacy Controls for Information Systems and Organizations," NIST Special Publication 800-53 Rev. 5, National Institute of

Standards and Technology, Gaithersburg, MD, USA, 2020, doi:
10.6028/NIST.SP.800-53r5.